SMK Manufacturing Analytics Dashboard

(End-to-End Manufacturing Analytics Solution)

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Tools & Technologies:

- Snowflake
- SQL
- Power BI
- DAX
- Dynamic RLS

Project Modules:

- ✅ Production Analytics
- ✅ Quality Analytics
- ✅ Maintenance Analytics
- ✅ Machine Performance Analytics

Key Features:

- Star Schema Data Model
- KPI Monitoring & Analysis
- Drilldown Reporting
- Executive Dashboard
- Dynamic Row-Level Security (RLS)

Project Summary:

The SMK Manufacturing Analytics Dashboard is a comprehensive business intelligence solution developed to monitor Production, Quality, Maintenance, and Machine Performance across manufacturing plants. The solution leverages Snowflake as the data warehouse and Power BI as the visualization platform to provide centralized reporting, KPI tracking, operational insights, and secure access through Dynamic Row-Level Security (RLS).

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1.Project Overview

The **SMK Manufacturing Analytics Dashboard** was developed to monitor and analyze Production, Quality, Maintenance, and Machine Performance across multiple manufacturing plants. The solution provides executives, regional managers, and plant managers with a centralized reporting platform to track key performance indicators (KPIs), identify operational issues, and make data-driven business decisions.

Project Description:

Manufacturing organizations generate large volumes of operational data from production processes, quality inspections, maintenance activities, and machine operations. Without a centralized reporting system, it becomes difficult for management teams to monitor performance and quickly identify operational bottlenecks.

To address this challenge, the SMK Manufacturing Analytics Dashboard was designed using **Snowflake** as the data warehouse and **Power BI** as the business intelligence platform. The solution integrates manufacturing data into a structured analytical model and presents business insights through interactive dashboards.

Key Functional Areas:

- Production Performance Monitoring
- Quality and Defect Analysis
- Downtime & Maintenance Tracking
- Machine-Level Performance Analysis
- Executive KPI Reporting
- Secure Data Access using Dynamic RLS (USERPRINCIPALNAME)

Expected Business Benefits:

- Improved visibility into manufacturing operations
- Faster identification of production issues
- Better monitoring of product quality
- Reduced downtime through maintenance analysis
- Enhanced decision-making using KPI-driven insights
- Secure and role-based access to business data

2.Business Problem

Manufacturing organizations often face challenges in monitoring operational performance across multiple plants due to the absence of a centralized reporting system. Business users require timely and accurate insights into production, quality, maintenance, and machine performance to support effective decision-making.

Key Challenges Identified:

- Low production efficiency affecting overall output.
- High defect rates impacting product quality and customer satisfaction.
- Excessive machine downtime causing production delays.
- Increasing maintenance costs reducing operational efficiency.
- Lack of centralized reporting for monitoring plant performance.

Business Requirement:

The organization needed a single analytics platform that could:

- Monitor manufacturing KPIs across all plants.
- Compare planned and actual production performance.
- Track quality metrics and defect trends.
- Analyze maintenance and downtime activities.
- Provide machine-level operational insights.
- Deliver executive-ready dashboards for business decision-making.

Proposed Solution:

To address these challenges, the SMK Manufacturing Analytics Dashboard was developed using Snowflake and Power BI to provide centralized reporting, KPI monitoring, drill-down analysis, and secure access through Dynamic Row-Level Security (RLS).

3. Project Objectives

The primary objective of the SMK Manufacturing Analytics Dashboard is to provide a centralized analytics platform that enables business users to monitor manufacturing operations, evaluate plant performance, and make data-driven decisions through interactive dashboards and KPI tracking.

Project Objectives:

1. Production Performance Monitoring

Monitor production activities across multiple manufacturing plants and compare actual production against planned targets to evaluate operational efficiency.

2. Production Analysis

Analyze production trends, production variance, achievement percentage, and plant-wise performance to identify areas of improvement.

3. Quality Monitoring

Track product quality performance using key metrics such as defect quantity, defect rate, and pass rate to improve manufacturing quality standards.

4. Maintenance & Downtime Analysis

Monitor maintenance activities, maintenance costs, and machine downtime to reduce operational disruptions and improve equipment reliability.

5. Machine Performance Analysis

Evaluate machine-level production output, downtime, and maintenance expenses to identify best-performing and underperforming machines.

6. Executive Reporting

Provide management and business stakeholders with interactive dashboards containing KPIs, trends, and insights for informed decision-making.

7. Secure Data Access

Implement Dynamic Row-Level Security (RLS) using USERPRINCIPALNAME() to ensure users can access only authorized data based on their roles and responsibilities.

Expected Outcomes

- Improved operational visibility across manufacturing plants.
- Faster identification of production and quality issues.
- Enhanced maintenance planning and downtime reduction.
- Better utilization of machines and resources.
- Secure and scalable reporting environment

4. Tools & Technologies

The SMK Manufacturing Analytics Dashboard was developed using modern data warehousing, business intelligence, and security technologies to enable efficient data management, analytics, and reporting.

Technology Stack

Category	Technology	Purpose
Data Warehousing	Snowflake	Centralized cloud data warehouse used to store and manage manufacturing data.
Data Querying	SQL	Used for data querying, validation, and analytical operations within Snowflake.
Data Modeling	Star Schema	Organizes fact and dimension tables to support analytical reporting.
Business Intelligence	Power BI Desktop	Used to create interactive dashboards, reports, and visual analytics.
Report Deployment	Power BI Service	Used to publish, share, and access reports online.
Business Calculations	DAX (Data Analysis Expressions)	Used to develop KPIs, measures, and business calculations.
Security & Access Control	Dynamic Row-Level Security (RLS)	Restricts data visibility based on user identity and assigned access rules.

Technology Overview:

Snowflake serves as the project's centralized cloud data warehouse, storing and managing manufacturing datasets used for reporting and analysis.

SQL is used for querying data, performing validations, and supporting analytical operations within the Snowflake environment.

Star Schema is implemented as the data modeling approach to establish relationships between fact and dimension tables, ensuring efficient reporting performance.

Power BI Desktop is used to design interactive dashboards, create visualizations, and develop analytical reports.

Power BI Service enables report publishing, online access, collaboration, and report sharing across stakeholders.

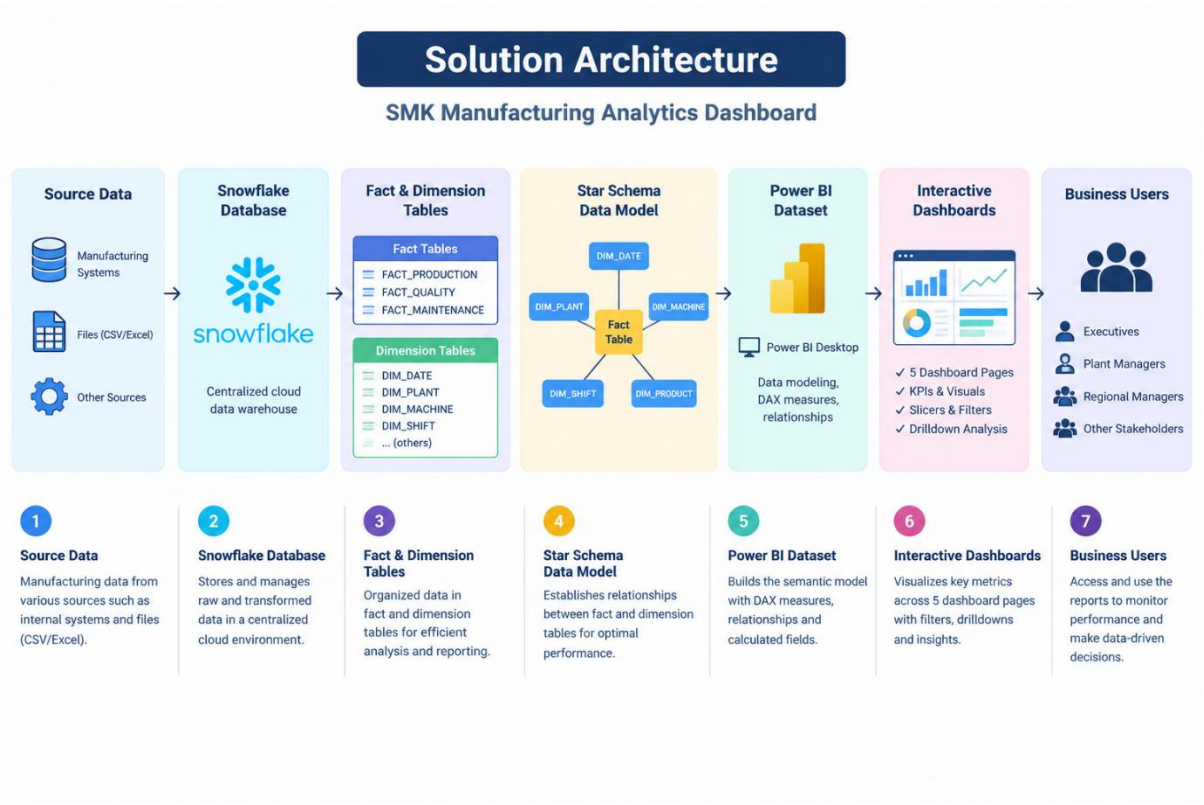
DAX (Data Analysis Expressions) is utilized to create KPIs, measures, and advanced business calculations that support decision-making.

Dynamic Row-Level Security (RLS) is implemented using USERPRINCIPALNAME() to ensure secure, role-based access to manufacturing data.

5. Solution Architecture

The SMK Manufacturing Analytics Dashboard follows a structured analytics architecture that enables data collection, storage, transformation, reporting, and secure business consumption. The architecture integrates Snowflake for data warehousing and Power BI for visualization and reporting.

Architecture Flow:



Architecture Components:

1.Source Data

Manufacturing data related to production, quality, maintenance, and machine performance serves as the source for the analytics solution.

2. Snowflake Database

All manufacturing datasets are stored and managed within Snowflake, providing a scalable and centralized cloud-based data warehouse environment.

3. Fact and Dimension Tables

Data is organized into Fact and Dimension tables to support efficient analytical reporting and simplify business analysis.

Fact Tables:	Dimension Tables:
• FACT_MAINTENANCE • FACT_DOWNTIME • FACT_PRODUCTION • FACT_PRODUCTION_TARGET • FACT_QUALITY	• DIM_DATE • DIM_PLANT • DIM_MACHINE • DIM_PRODUCT • DIM_SHIFT • DIM_LINE • DIM_USER_ACCESS

4. Star Schema Data Model

A Star Schema model is implemented to establish relationships between fact and dimension tables. This design improves report performance, simplifies DAX calculations, and enhances scalability.

5. Power BI Dataset

The modeled data is imported into Power BI, where relationships, measures, KPIs, and business calculations are created using DAX.

6. Interactive Dashboards

Power BI dashboards provide visual insights into production, quality, maintenance, downtime, and machine performance metrics.

7. Business Users

Executives, Regional Managers, and Plant Managers access dashboards based on their assigned permissions through Dynamic Row-Level Security (RLS).

Benefits of the Architecture

- Centralized data management.
- Scalable cloud-based analytics solution.
- Improved dashboard performance through Star Schema.
- Efficient KPI and business metric calculations.
- Secure access using Dynamic RLS.
- Self-service reporting for business users

6. Data Model

The SMK Manufacturing Analytics Dashboard follows a dimensional data modeling approach using a Star Schema design. The model consists of Fact Tables that store transactional and measurable data and Dimension Tables that provide descriptive business information for analysis and reporting.

Purpose of the Data Model

The data model was designed to:

- Support efficient analytical reporting.
- Improve dashboard performance.
- Simplify business analysis through dimensions.
- Enable accurate KPI calculations using DAX measures.
- Establish a scalable architecture for future enhancements.

Data Model Structure

The solution consists of:

Fact Tables:

Fact tables store quantitative business data generated from manufacturing operations.

- FACT_MAINTENANCE
- FACT_DOWNTIME
- FACT_PRODUCTION
- FACT_PRODUCTION_TARGET
- FACT_QUALITY

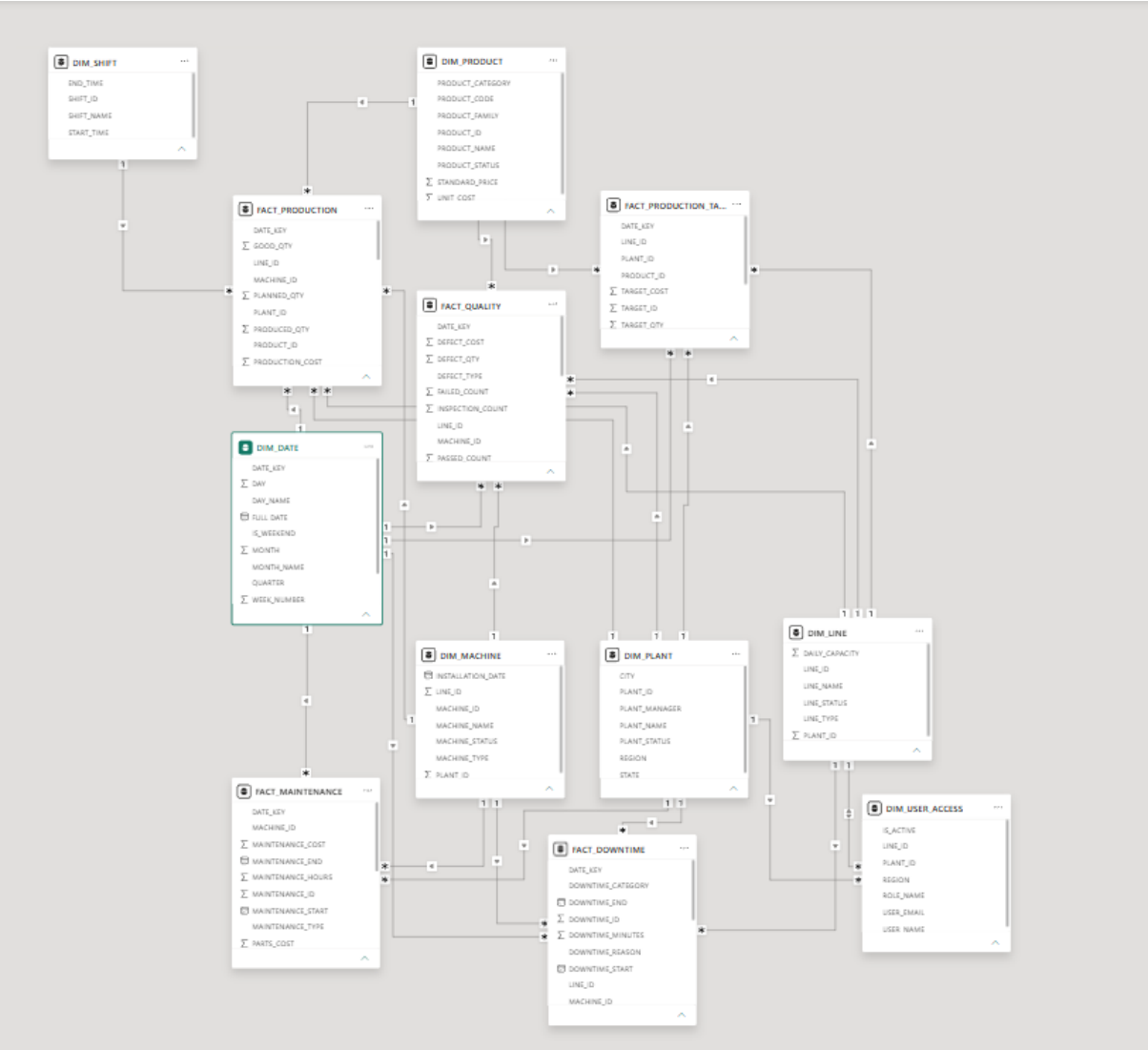
Dimension Tables: Dimension tables provide descriptive attributes used for filtering, grouping, and business analysis.

- DIM_DATE
- DIM_PLANT
- DIM_MACHINE
- DIM_PRODUCT
- DIM_SHIFT
- DIM_LINE
- DIM_USER_ACCESS

Data Model Benefits

- Faster report performance.
- Simplified relationship management.
- Consistent KPI calculations.
- Better scalability and maintainability.
- Improved user experience for business reporting. [smk document stech | PDF]

Model Overview



The dimensional model enables users to analyze production, quality, maintenance, and machine performance data from multiple business perspectives while maintaining high reporting efficiency.

6.1 Fact Tables:

Fact tables store transactional and measurable data generated from manufacturing operations. These tables contain business metrics that are analyzed across different dimensions such as Plant, Date, Machine, and Defect Type.

Fact Table	Description
FACT_PRODUCTION	Stores actual production data including produced quantity, production cost, and related production metrics.
FACT_PRODUCTION_TARGET	Stores planned production targets used for target vs actual performance analysis.
FACT_QUALITY	Stores quality inspection and defect-related information used for quality analysis and defect monitoring.
FACT_MAINTENANCE	Stores maintenance events, maintenance costs, and maintenance activity details.
FACT_DOWNTIME	Stores downtime records including downtime duration, category, and machine-related downtime information.

FACT_PRODUCTION

Purpose:
Stores actual production performance data generated from manufacturing operations.

Key Business Usage:

- Total Production Analysis
- Production Trend Analysis
- Production by Plant
- Production by Machine
- Production Cost Analysis

FACT_PRODUCTION_TARGET

Purpose:
Stores planned production targets used to measure production achievement and variance.

Key Business Usage:

- Planned vs Actual Comparison
- Achievement %
- Production Variance
- Target Performance Monitoring

FACT_QUALITY

Purpose:
Stores inspection results, defect information, and quality-related manufacturing metrics.

Key Business Usage:

- Defect Analysis
- Defect Rate Calculation
- Pass Rate Analysis
- Quality Performance Monitoring

FACT_MAINTENANCE

Purpose:
Stores maintenance activity records and associated maintenance costs.

Key Business Usage:

- Maintenance Cost Analysis
- Maintenance Event Tracking
- Maintenance Type Analysis
- Plant-wise Maintenance Monitoring

FACT_DOWNTIME

Purpose:
Stores machine and production downtime information.

Key Business Usage:

- Total Downtime Analysis
- Downtime by Plant
- Downtime by Machine
- Downtime Category Analysis
- Operational Efficiency Monitoring

6.2 Dimension Tables

Dimension tables store descriptive business attributes that provide context to transactional data stored in fact tables. These tables support filtering, drill-down analysis, grouping, and reporting across multiple business perspectives.

Dimension Table	Description
DIM_DATE	Stores date-related attributes used for time-based analysis and reporting.
DIM_PLANT	Stores plant information used for plant-wise performance analysis.
DIM_MACHINE	Stores machine details used for machine-level reporting and analysis.
DIM_PRODUCT	Stores product-related information used for production and quality analysis.
DIM_SHIFT	Stores shift information used for shift-wise performance monitoring.
DIM_LINE	Stores production line details used for line-level operational analysis.
DIM_USER_ACCESS	Stores user access information used to implement Dynamic Row-Level Security (RLS).

DIM_DATE

Purpose: Provides calendar and time-related attributes for trend and period analysis.

Business Usage:

- Daily Analysis
- Monthly Reporting
- Quarterly Analysis
- Year-over-Year Comparison

DIM_PLANT

Purpose: Stores manufacturing plant information.

Business Usage:

- Plant Performance Analysis
- Plant-wise Production Reporting
- Quality Analysis by Plant
- Maintenance Analysis by Plant

DIM_MACHINE

Purpose: Stores machine-related information.

Business Usage:

- Machine Performance Analysis
- Downtime by Machine
- Production by Machine
- Maintenance Cost Analysis

DIM_PRODUCT

Purpose: Stores product information used for production and quality analysis.

Business Usage:

- Product-wise Production Analysis
- Product Quality Monitoring
- Defect Analysis by Product
- Product Performance Tracking

DIM_SHIFT

Purpose: Stores shift-related information for operational monitoring.

Business Usage:

- Shift-wise Production Analysis
- Shift Efficiency Monitoring
- Shift Quality Analysis
- Shift Downtime Analysis

DIM_LINE

Purpose: Stores production line information.

Business Usage:

- Production by Line
- Line Efficiency Analysis
- Quality by Line
- Downtime by Line

DIM_USER_ACCESS

Purpose: Stores user details and plant assignments used for Dynamic Row-Level Security (RLS).

Business Usage:

- Role-Based Data Access
- Plant-Level Security
- User Authorization Management
- Dynamic RLS Implementation using USERPRINCIPALNAME()

Benefits of Dimension Tables

- Enable filtering and drill-down analysis.
- Provide business context to fact table data.
- Improve report usability and navigation.
- Support Dynamic RLS implementation.
- Enhance analytical flexibility across dashboards.

7. Star Schema Design

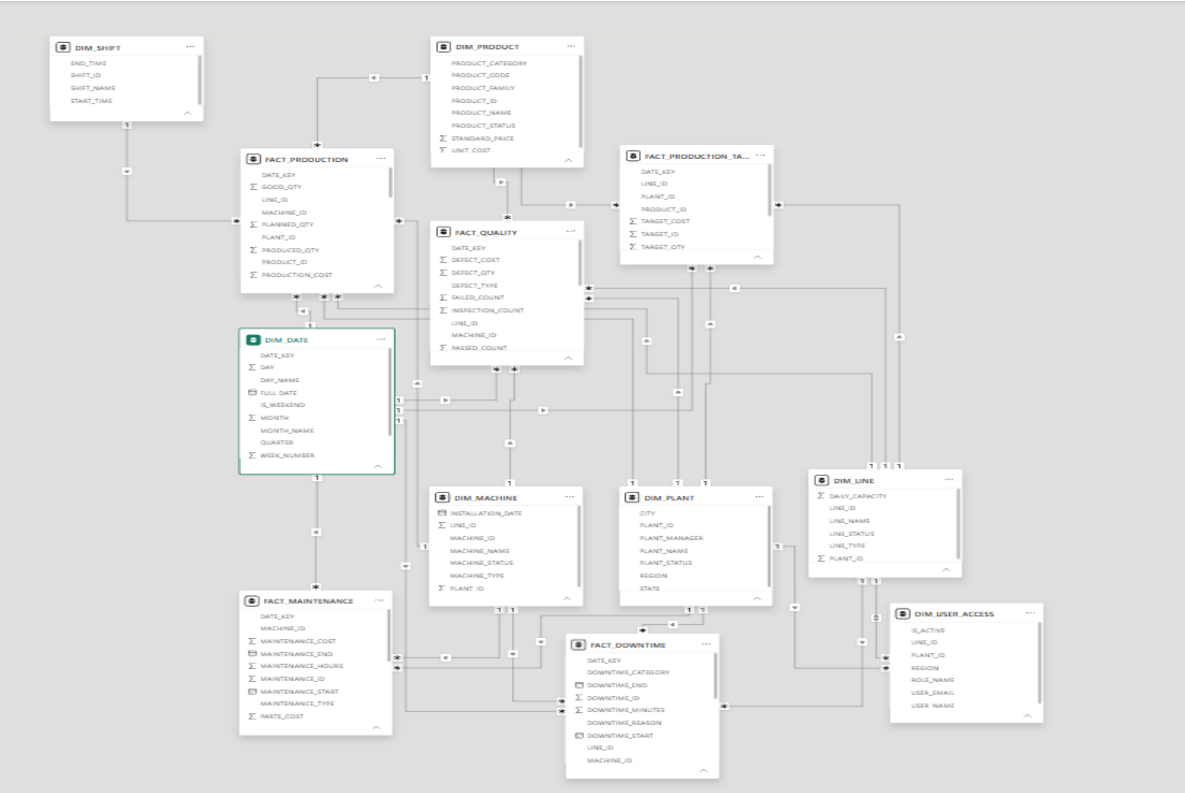
The SMK Manufacturing Analytics Dashboard follows a **Star Schema** data modeling approach, which is widely used in Business Intelligence and Data Warehousing projects. This design consists of centralized Fact Tables connected to multiple Dimension Tables, providing efficient query performance and simplified reporting.

Star Schema Overview:

In the data model, Fact Tables store transactional and measurable business data, while Dimension Tables provide descriptive information used for filtering, grouping, and drill-down analysis.

Fact Tables:	**Dimension Tables:**
• FACT_MAINTENANCE • FACT_DOWNTIME • FACT_PRODUCTION • FACT_PRODUCTION_TARGET • FACT_QUALITY	• DIM_DATE • DIM_PLANT • DIM_MACHINE • DIM_PRODUCT • DIM_SHIFT • DIM_LINE • DIM_USER_ACCESS

Data Model Representation:



Benefits of Star Schema Design

- Simplifies data relationships and model maintenance.
- Improves report and dashboard performance.
- Enables faster DAX calculations.
- Supports scalable reporting architecture.
- Provides easier navigation and analysis for business users.
- Improves drill-down and filtering capabilities across dashboards.

Why Star Schema Was Used

The Star Schema structure was selected because it provides a clean separation between transactional data and descriptive business attributes. This approach improves analytical performance while making the data model easier to understand, maintain, and extend for future reporting requirements.

8. Major DAX Measures

The dashboard utilizes DAX (Data Analysis Expressions) measures to calculate Key Performance Indicators (KPIs) and provide meaningful business insights. These measures support Production, Quality, Maintenance, and Downtime analysis across manufacturing plants.

8.1 Production Measures

Total Produced Quantity

Purpose: Calculates the total quantity produced across all manufacturing plants.

DAX

```
Total Produced Qty = SUM(FACT_PRODUCTION[PRODUCED_QTY])
```

Total Planned Quantity

Purpose: Calculates the total planned production quantity.

DAX

```
Total Planned Qty =SUM(FACT_PRODUCTION[PLANNED_QTY])
```

Total Rejected Quantity

Purpose: Calculates the total quantity of rejected products.

DAX

```
Total Rejected Qty =SUM(FACT_PRODUCTION[REJECTED_QTY])
```

Production Achievement %

Purpose: Measures the percentage of actual production achieved against the planned target.

DAX

```
Achievement % =DIVIDE([Total Produced Qty], [Total Planned Qty],0 ) * 100
```

Production Variance

Purpose: Calculates the difference between actual production and planned production target.

DAX

```
Production Variance =[Total Produced Qty] - [Total Planned Qty]
```

Business Usage:

- Measures production gap against target.
- Identifies overachievement or underachievement.
- Supports operational performance analysis.
- Helps management evaluate production efficiency.

Achievement Rank

Purpose: Ranks manufacturing plants based on their production achievement percentage.

DAX

```
Achievement Rank =RANKX(
ALL(DIM_PLANT[PLANT_NAME]),[Achievement %],,DESC,DENSE)
```

Business Usage:

- Identify top-performing plants.
- Compare production efficiency across plants.
- Support executive performance evaluation.

Achievement Rank Color

Purpose: Applies dynamic color formatting to identify the highest and lowest performing plants.

DAX

```
Achievement Rank Color =
VAR MinDefect =MINX(ALL(DIM_PLANT[PLANT_NAME]),[Achievement %])
VAR MaxDefect =MAXX(ALL(DIM_PLANT[PLANT_NAME]),[Achievement %])
RETURN
SWITCH(
TRUE(), [Achievement %] = MaxDefect, "#28A745", [Achievement %] = MinDefect, "#DC3545",
"#4B6E78" )
```

Color Logic:

- Green → Best Performing Plant
- Red → Lowest Performing Plant
- Teal → Other Plants

Production Drilldown Color

Purpose: Applies conditional formatting during drill-down analysis at Line level.

DAX

```
Production Drilldown Color =IF( ISINSCOPE(DIM_LINE[LINE_NAME]),
VAR MinProd =MINX(ALLSELECTED(DIM_LINE[LINE_NAME]),[Total Produced Qty])
VAR MaxProd =MAXX(ALLSELECTED(DIM_LINE[LINE_NAME]),[Total Produced Qty])
RETURN
SWITCH(
TRUE(), [Total Produced Qty] = MaxProd, "#28A745", [Total Produced Qty] = MinProd, "#DC3545", "#4B6E78"),  "#4B6E78" )
```

Business Usage:

- Highlights best-performing production lines.
- Identifies low-performing lines instantly during drill-down analysis.

Production KPI Summary

Measure	Business Purpose
Total Produced Qty	Tracks the total quantity produced across all manufacturing plants.
Total Planned Qty	Measures the planned production target for comparison and performance evaluation.
Total Rejected Qty	Monitors the quantity of rejected products during manufacturing.
Production Variance	Identifies the gap between actual production and planned production targets.
Achievement %	Measures how effectively production targets are achieved.
Achievement Rank	Ranks plants based on production achievement performance.
Achievement Rank Color	Highlights the best and worst performing plants using conditional formatting.
Production Drilldown Color	Highlights top and bottom performing production lines during drill-down analysis.

8.2 Quality Measures

The Quality Analytics dashboard uses DAX measures to monitor inspections, pass rate, defect rate, and overall product quality performance across manufacturing plants.

Total Inspections

Purpose: Calculates the total number of inspections performed.

DAX

```
Total Inspections = SUM(FACT_QUALITY[INSPECTION_COUNT])
```

Total Passed

Purpose: Calculates the total number of products that passed inspection.

DAX

```
Total Passed =
SUM(FACT_QUALITY[PASSED_COUNT])
```

Total Failed

Purpose: Calculates the total number of products that failed inspection.

DAX

```
Total Failed =SUM(FACT_QUALITY[FAILED_COUNT])
```

Total Defect Quantity

Purpose: Calculates the total defective quantity identified during inspections.

DAX

```
Total Defect Qty =SUM(FACT_QUALITY[DEFECT_QTY])
```

Pass Rate %

Purpose: Measures the percentage of products that successfully passed quality inspection.

DAX

```
Pass Rate % =
DIVIDE([Total Passed],[Total Inspections],0)
```

Defect Rate %

Purpose: Measures the percentage of defective products identified during inspections.

```
DAX
Defect Rate % =
DIVIDE(
[Total Defect Qty],
[Total Inspections],
0
)
```

Defect Status

Purpose: Classifies plant quality performance based on defect rate thresholds.

```
DAX
Defect Status =
SWITCH(
TRUE(),
[Defect Rate %] <= 0.02, "Excellent",
[Defect Rate %] <= 0.05, "Good",
[Defect Rate %] <= 0.08, "Average",
"Needs Improvement"
)
```

Quality Categories:

Defect Rate	Status
≤ 2%	Excellent
≤ 5%	Good
≤ 8%	Average
> 8%	Needs Improvement

Defect Percentage Rank

Purpose: Ranks plants based on defect rate performance.

DAX

```
Defect Percentage Rank =
RANKX(
ALL(DIM_PLANT[PLANT_NAME]),[Defect Rate %],
,ASC,DENSE)
```

Note: Lower defect rate receives a better rank.

Defect Percentage Rank Color

Purpose: Highlights the best and worst performing plants based on defect rate.

DAX

```
Defect Percentage Rank Color =
VAR MinDefect =MINX(ALL(DIM_PLANT[PLANT_NAME]),[Defect Rate %] )
VAR MaxDefect =MAXX(ALL(DIM_PLANT[PLANT_NAME]),[Defect Rate %] )
RETURN
SWITCH(
TRUE(),
[Defect Rate %] = MinDefect, "#28A745",
[Defect Rate %] = MaxDefect, "#DC3545",
"#4B6E78"
)
```

Color Logic:

- Green → Lowest Defect Rate (Best Plant)
- Red → Highest Defect Rate (Worst Plant)
- Teal → Other Plants

Defect Drilldown Color

Purpose: Applies conditional formatting during drill-down analysis by defect type.

```
DAX

Defect Drilldown Color =

VAR CurrentValue = [Total Defect Qty]

VAR MinValue MINX(ALLSELECTED(FACT_QUALITY[DEFECT_TYPE]),[Total Defect Qty])

VAR MaxValue =MAXX(ALLSELECTED(FACT_QUALITY[DEFECT_TYPE]),[Total Defect Qty])

RETURN

SWITCH(

TRUE(),

CurrentValue = MaxValue, "#DC3545",

CurrentValue = MinValue, "#28A745",

"#4B6E78"  )
```

Quality KPI Summary

Measure	Business Purpose
Total Inspections	Tracks total inspection activities conducted.
Total Passed	Measures successfully passed products.
Total Failed	Measures failed products during inspection.
Total Defect Qty	Tracks defective product quantity.
Pass Rate %	Measures quality success rate.
Defect Rate %	Measures defect occurrence percentage.
Defect Status	Classifies quality performance.
Defect Percentage Rank	Ranks plants by quality performance.
Defect Percentage Rank Color	Highlights best and worst plants.
Defect Drilldown Color	Highlights best and worst defect categories during drilldown.

8.3 Maintenance & Downtime Measures

The Maintenance and Downtime Analytics dashboards use DAX measures to monitor maintenance expenses, downtime duration, machine performance, and equipment reliability.

Maintenance Measures

Total Maintenance Cost

Purpose: Calculates total maintenance expenditure.

```
Total Maintenance Cost = SUM(FACT_MAINTENANCE[MAINTENANCE_COST])
```

Maintenance Events

Purpose: Counts total maintenance activities performed.

```
Maintenance Events = DISTINCTCOUNT(FACT_MAINTENANCE[MAINTENANCE_ID])
```

Cost Per Event

Purpose: Calculates average maintenance cost per maintenance activity.

```
Cost Per Event = DIVIDE(    [Total Maintenance Cost],    [Maintenance Events]  )
```

Total Maintenance Hours

Purpose: Calculates total maintenance hours consumed.

```
Total Maintenance Hours =SUM(FACT_MAINTENANCE[MAINTENANCE_HOURS])
```

Avg Maintenance Cost

Purpose: Calculates average maintenance cost per maintenance event.

```
Avg Maintenance Cost =
DIVIDE(   [Total Maintenance Cost],    [Maintenance Events],    0  )
```

Maintenance Status

Purpose: Classifies maintenance performance based on average maintenance duration.

```
Maintenance Status =
SWITCH(    TRUE(),
   [Average Maintenance Hours] <= 2, "Excellent",
   [Average Maintenance Hours] <= 4, "Good",
   [Average Maintenance Hours] <= 8, "Average",
   "Needs Improvement"  )
```

Maintenance Color

Purpose: Highlights plants with the lowest and highest maintenance costs.

```
Maintenance Color =
VAR MinDowntime =
  MINX(
    ALL(DIM_PLANT[PLANT_NAME]),
    [Total Maintenance Cost]
  )
VAR MaxDowntime =
  MAXX(
    ALL(DIM_PLANT[PLANT_NAME]),
    [Total Maintenance Cost]
  )
RETURN
SWITCH(
  TRUE(),
  [Total Maintenance Cost] = MinDowntime, "#28A745",
  [Total Maintenance Cost] = MaxDowntime, "#DC3545",
  "#4B6E78"
)
```

Downtime Measures

Total Downtime (Minutes)

Purpose: Calculates total downtime across all machines and plants.

```
Total Downtime(Minutes) =SUM(FACT_DOWNTIME[DOWNTIME_MINUTES])
```

Total Downtime (Hours)

Purpose: Displays downtime in hours format.

```
Total Downtime(Hrs) =
FORMAT(    [Total Downtime Hrs measure],    "#,##0" ) & " Hrs"
```

Downtime Events

Purpose: Counts total downtime incidents.

```
Downtime Events = DISTINCTCOUNT(FACT_DOWNTIME[DOWNTIME_ID] )
```

Downtime Per Event

Purpose: Calculates average downtime generated per maintenance event.

```
Downtime / Event (Hrs) = DIVIDE(    [Total Downtime(Minutes)],   [Maintenance Events]  )
```

Downtime Color

Purpose: Highlights plants with lowest and highest downtime.

```
Downtime Color =
VAR MinDowntime =
   MINX(       ALL(DIM_PLANT[PLANT_NAME]),      [Total Downtime(Minutes)]   )
VAR MaxDowntime =
   MAXX(    ALL(DIM_PLANT[PLANT_NAME]),   [Total Downtime(Minutes)]   )
RETURN
SWITCH(
   TRUE(),
   [Total Downtime(Minutes)] = MinDowntime, "#28A745",
   [Total Downtime(Minutes)] = MaxDowntime, "#DC3545",
   "#4B6E78"
)
```

Machine Performance Measures

Best Machine

Purpose: Identifies the machine with the highest production achievement.

```
Best Machine =
VAR TopMachine =
   TOPN(
      1,
      ALL(DIM_MACHINE[MACHINE_TYPE]),
      [Achievement %],
```

```
    DESC
  )
RETURN
CONCATENATEX(
  TopMachine,
  DIM_MACHINE[MACHINE_TYPE],
  ", "
)
```

Best Machine Achievement %

```
Best Machine Achievement % =
MAXX(   ALL(DIM_MACHINE[MACHINE_TYPE]),  [Achievement %] )
```

Worst Machine

Purpose: Identifies the machine with the highest downtime.

```
Worst Machine =
VAR WorstMachine =
  TOPN(
    1,
    ALL(DIM_MACHINE[MACHINE_TYPE]),
    [Total Downtime(Minutes)],
    DESC   )
RETURN
CONCATENATEX(
  WorstMachine,
  DIM_MACHINE[MACHINE_TYPE],
  ", "
)
```

Worst Machine Downtime

```
Worst Machine Downtime =
MAXX(  ALL(DIM_MACHINE[MACHINE_TYPE]),   [Total Downtime(Minutes)]  )
```

Worst Machine Downtime Hours

```
Worst Machine Downtime Hrs = DIVIDE(  [Worst Machine Downtime],  60 )
```

Maintenance & Downtime KPI Summary

Measure	Business Purpose
Total Maintenance Cost	Tracks overall maintenance expenditure
Maintenance Events	Counts maintenance activities
Cost Per Event	Measures average maintenance cost
Total Maintenance Hours	Tracks maintenance effort
Avg Maintenance Cost	Evaluates maintenance efficiency
Maintenance Status	Classifies maintenance performance
Maintenance Color	Highlights best/worst maintenance cost plants
Total Downtime (Minutes)	Tracks downtime duration
Downtime Events	Counts downtime incidents
Downtime Per Event	Measures average downtime impact
Downtime Color	Highlights best/worst downtime plants
Best Machine	Identifies top-performing machine
Best Machine Achievement %	Measures best machine efficiency
Worst Machine	Identifies highest downtime machine
Worst Machine Downtime	Measures downtime impact
Worst Machine Downtime Hrs	Downtime displayed in hours

Executive Overview Dashboard

9.1 Executive Overview

Purpose

The Executive Overview dashboard provides a consolidated view of key manufacturing KPIs across all plants. It enables management to quickly assess production performance, product quality, maintenance expenditure, and operational downtime through a single reporting page.

Key KPIs

- Total Produced Quantity
- Total Maintenance Cost
- Achievement %
- Defect Rate %
- Total Downtime (Hours)

Key Visualizations

- Total Produced Quantity by Plant
- Total Maintenance Cost by Plant
- Achievement % by Plant
- Defect Rate % by Plant
- Total Downtime (Hours) by Plant

Filters Available

- Date Range
- Region
- Plant Name

Business Insights

- Identifies the highest and lowest performing plants.
- Monitors achievement against production targets.
- Compares defect rates across manufacturing plants.
- Tracks maintenance cost distribution.
- Highlights plants with excessive downtime.

Business Value

The Executive Overview dashboard provides a quick summary of operational performance and enables executives to monitor critical KPIs without navigating through multiple reports.



9.2 Production Performance

Purpose

The Production Performance dashboard focuses on analyzing production output, target achievement, production trends, and operational efficiency across manufacturing plants and production lines.

Key KPIs

- Total Planned Quantity
- Total Produced Quantity
- Achievement %
- Production Variance
- Total Production Cost
- Total Rejected Quantity

Key Visualizations

- Production by Line
- Produced Quantity by Plant
- Planned vs Actual Production by Plant
- Current Year vs Previous Year Production Trend
- Production Performance Matrix

Drilldown Analysis

Production Qty by Plant	**Planned vs Actual Production by Plant**
Plant ↓ Line ↓ Year ↓ Quarter ↓ Month	Plant ↓ Year
Purpose: Allows users to navigate from high-level plant performance to detailed monthly production analysis.	**Purpose** Enables year-wise analysis of production targets and achievements to track performance trends over time.

Key Measures Used

- Total Produced Qty
- Total Planned Qty
- Total Rejected Qty
- Production Variance
- Achievement %
- Achievement Rank
- Achievement Rank Color

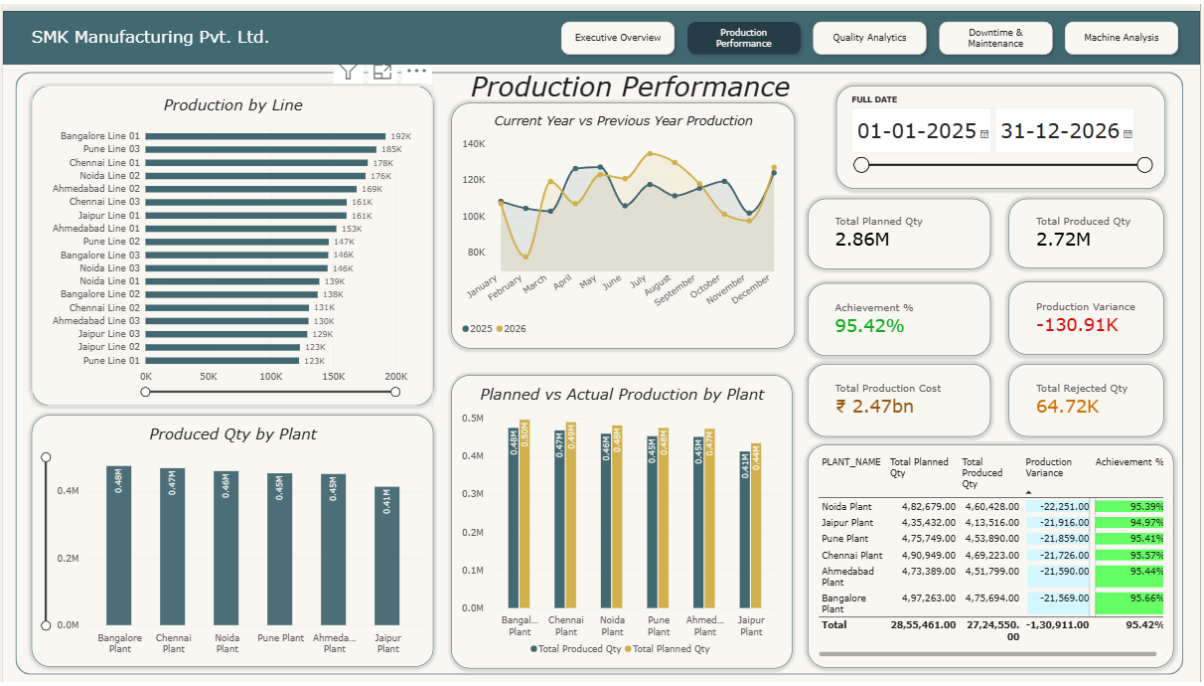
- Production Drilldown Color

Business Insights

- Compares planned production against actual production.
- Identifies production gaps and operational bottlenecks.
- Evaluates plant-wise production efficiency.
- Analyzes production trends across different years.
- Supports detailed drill-down analysis from plant level to monthly production performance.

Business Value

The Production Performance dashboard helps management monitor production efficiency, identify underperforming production lines, and improve resource utilization through data-driven decision-making.



9.3 Quality Analytics

Purpose

The Quality Analytics dashboard is designed to monitor product quality, defect trends, inspection results, and plant-wise quality performance. It helps identify quality issues and supports continuous improvement initiatives.

Key KPIs

- Total Inspections
- Total Passed
- Total Failed
- Total Defect Qty
- Pass Rate %
- Defect Rate %
- Defect Status

Key Visualizations

- Defects by Plant
- Defects by Type
- Defect Rate Trend Over Time

Drilldown Analysis

Defects by Plant

Plant

↓

Defect Type

↓

Year

Purpose:

- Allows users to identify defect quantity by plant and drill down into specific defect types.
- Enables year-wise analysis of defect trends to monitor quality performance over time.

Key Measures Used

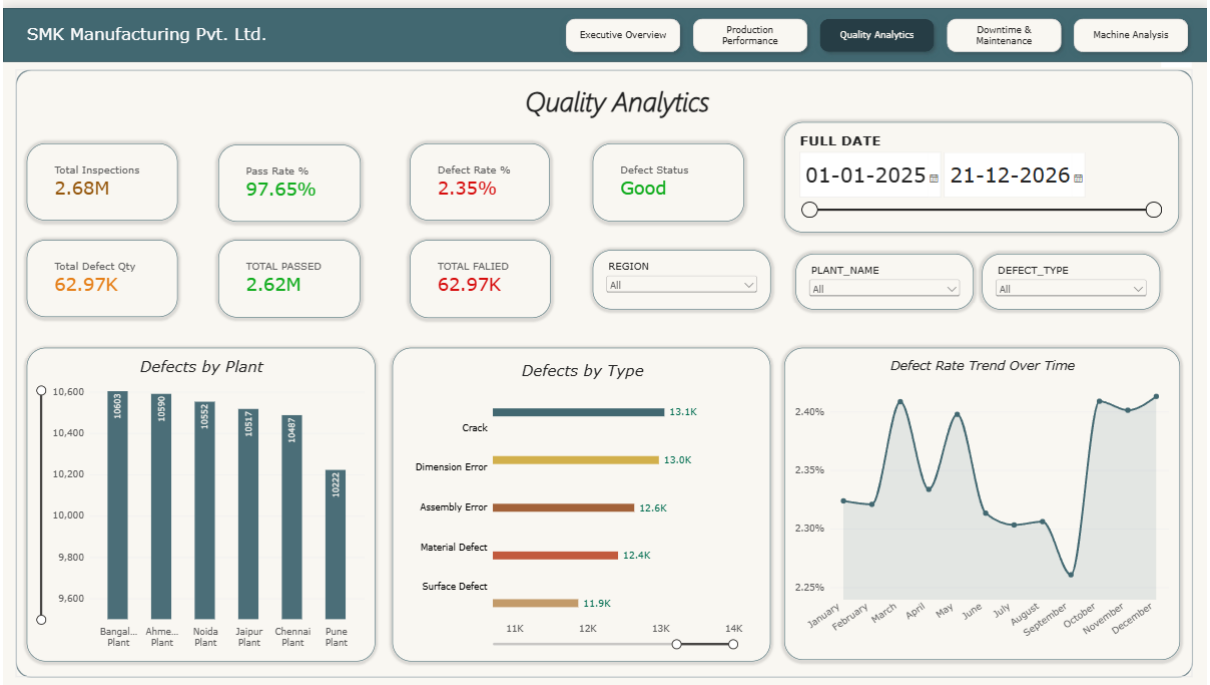
- Total Inspections
- Total Passed
- Total Failed
- Total Defect Qty
- Pass Rate %
- Defect Rate %
- Defect Status
- Defect Percentage Rank
- Defect Percentage Rank Color
- Defect Drilldown Color

Business Insights

- Identifies plants with the highest defect quantities.
- Highlights major defect categories affecting product quality.
- Supports root-cause analysis through defect-type drilldowns.

Business Value

The Quality Analytics dashboard helps quality teams monitor manufacturing quality, reduce defect rates, improve pass rates, and support data-driven quality improvement initiatives.



9.4 Downtime & Maintenance Analytics

Purpose

The Downtime & Maintenance Analytics dashboard provides insights into equipment downtime, maintenance costs, maintenance activities, and machine reliability across manufacturing plants. It helps identify operational inefficiencies and optimize maintenance performance.

Key KPIs

- Total Maintenance Cost
- Cost Per Event
- Total Downtime (Hours)
- Downtime Per Event (Hours)
- Total Maintenance Hours
- Maintenance Events
- Downtime Events
- Maintenance Status

Key Visualizations

- Total Downtime (Hours) by Plant
- Total Maintenance Cost by Plant
- Downtime by Category
- Maintenance by Type

Drilldown Analysis

Total Downtime (Hours) by Plant	**Total Maintenance Cost by Plant**
Plant ↓ Year	Plant ↓ Machine Type ↓ Year

Purpose: • Allows users to analyze downtime hours at the plant level and drill down to yearly downtime trends. • Helps identify changes in equipment availability and operational efficiency over different year	**Purpose:** • Allows users to analyze maintenance costs by plant and drill down to machine-level maintenance expenses. • Enables year-wise maintenance cost tracking to identify high-cost machines and optimize maintenance strategies.

Key Measures Used

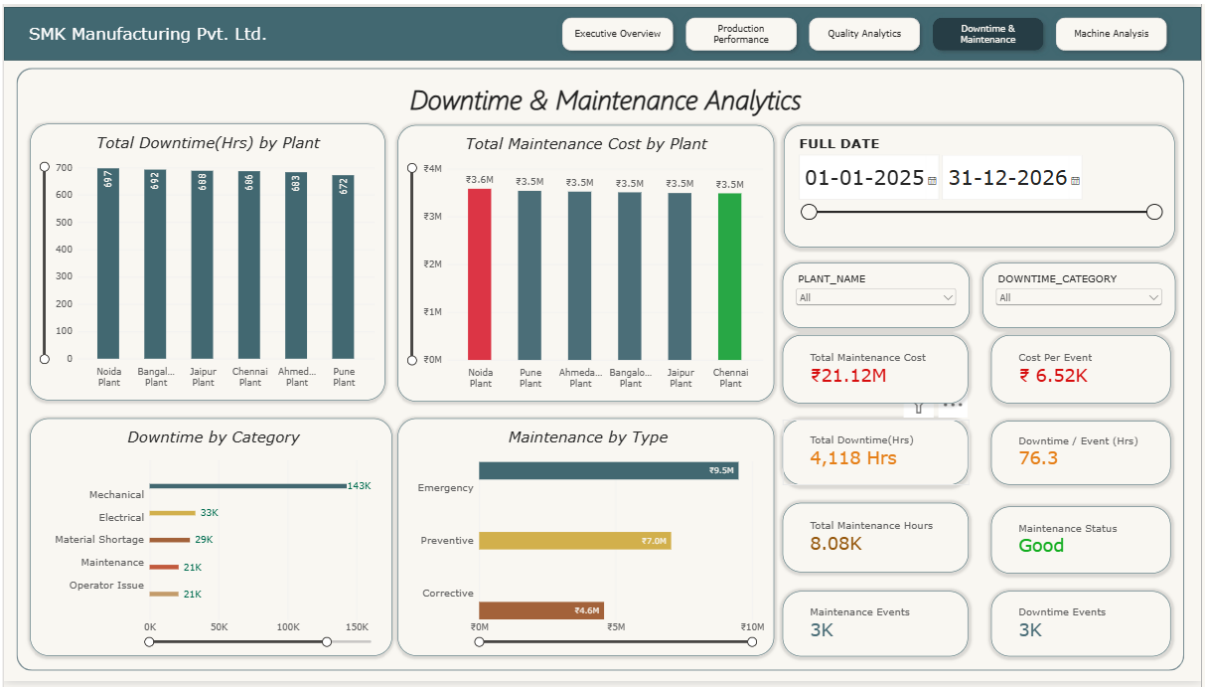
- Total Maintenance Cost
- Maintenance Events
- Cost Per Event
- Total Maintenance Hours
- Avg Maintenance Cost
- Maintenance Status
- Maintenance Color
- Total Downtime (Minutes)
- Total Downtime (Hours)
- Downtime Events
- Downtime / Event (Hours)
- Downtime Color
- Best Machine
- Worst Machine
- Worst Machine Downtime
- Worst Machine Downtime Hours

Business Insights

- Identifies plants with the highest downtime.
- Highlights plants incurring the highest maintenance costs.
- Analyzes downtime categories causing production losses.
- Evaluates maintenance effectiveness through cost and event metrics.

Business Value

The Downtime & Maintenance Analytics dashboard enables proactive maintenance management, reduces operational downtime, improves equipment reliability, and supports cost optimization initiatives.



9.5 Machine Analysis

Purpose

The Machine Analysis dashboard provides detailed insights into machine-level production, maintenance costs, downtime, and operational efficiency. It helps identify the best-performing and underperforming machines across manufacturing plants.

Key KPIs

- Total Produced Quantity
- Total Downtime (Hours)
- Total Maintenance Cost
- Average Maintenance Cost
- Best Machine
- Worst Machine

Key Visualizations

- Produced Quantity by Machine Type
- Downtime (Hours) by Machine Type
- Maintenance Cost by Machine Type
- Machine Performance Matrix

Key Measures Used

- Total Produced Qty
- Total Downtime (Minutes)
- Total Downtime (Hours)
- Total Maintenance Cost
- Avg Maintenance Cost
- Best Machine
- Best Machine Achievement %
- Worst Machine
- Worst Machine Downtime
- Worst Machine Downtime Hours

Business Insights

- Identifies machines contributing the highest production output.
- Highlights machines with excessive downtime.
- Compares maintenance costs across machine types.
- Evaluates machine performance and operational efficiency.
- Supports maintenance planning and machine utilization decisions.

Business Value

The Machine Analysis dashboard enables organizations to monitor machine performance, reduce downtime, optimize maintenance costs, and improve overall manufacturing efficiency.

Machine Performance Indicators

Best Machine

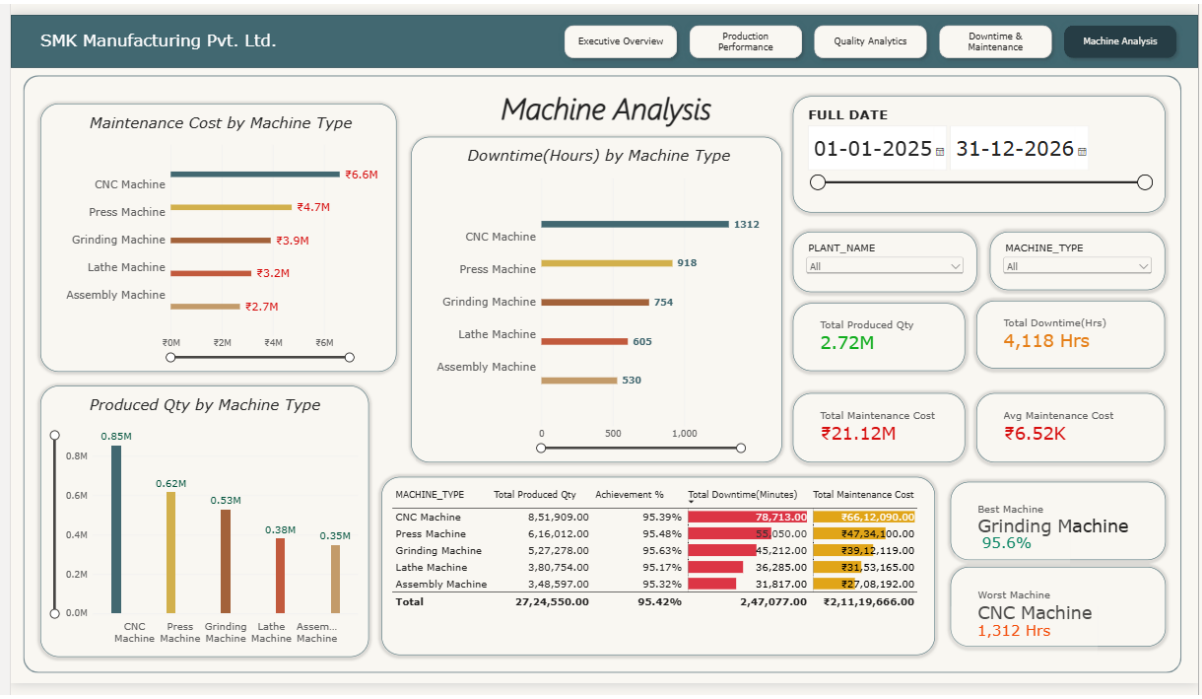
Purpose: Identifies the machine with the highest production achievement percentage.

Worst Machine

Purpose: Identifies the machine with the highest downtime and maintenance impact

Dashboard Highlights

- CNC Machine contributes the highest production output.
- Machine-wise downtime analysis helps identify performance bottlenecks.
- Maintenance cost analysis supports cost optimization initiatives.
- Best and Worst Machine KPIs provide quick operational insights.
- Machine-level analytics improve equipment utilization and planning.



10. Dynamic Row-Level Security (RLS)

Purpose

Dynamic Row-Level Security (RLS) was implemented to ensure that users can access only the data relevant to their assigned role and responsibilities. The security model dynamically filters dashboard data based on the logged-in user's email using the USERPRINCIPALNAME() function.

Security Roles Implemented

1. CEO

Access Level:

- Access to all manufacturing plants.
- Access to all regions.
- Complete visibility across the organization.

RLS Logic:

```
DAX
COUNTROWS(
FILTER(
DIM_USER_ACCESS,
DIM_USER_ACCESS[USER_EMAIL] = USERPRINCIPALNAME()
&&
DIM_USER_ACCESS[ROLE_NAME] = "CEO"
&&
DIM_USER_ACCESS[IS_ACTIVE] = TRUE()
)
) > 0
```

2. Plant Manager

Access Level:

- Access restricted to assigned plant only.
- Cannot view data from other plants.

RLS Logic:

```
DAX
VAR PlantID =
LOOKUPVALUE(
DIM_USER_ACCESS[PLANT_ID],
DIM_USER_ACCESS[USER_EMAIL], USERPRINCIPALNAME(),
DIM_USER_ACCESS[ROLE_NAME], "Plant Manager"
)

RETURN
DIM_PLANT[PLANT_ID] = PlantID
```

3. Regional Manager

Access Level:

- Access restricted to assigned region.
- Can view all plants within the assigned region.

RLS Logic:

```
DAX
VAR UserRegion =
LOOKUPVALUE(
DIM_USER_ACCESS[REGION],
DIM_USER_ACCESS[USER_EMAIL], USERPRINCIPALNAME(),
DIM_USER_ACCESS[ROLE_NAME], "Regional Manager",
DIM_USER_ACCESS[IS_ACTIVE], TRUE()
)

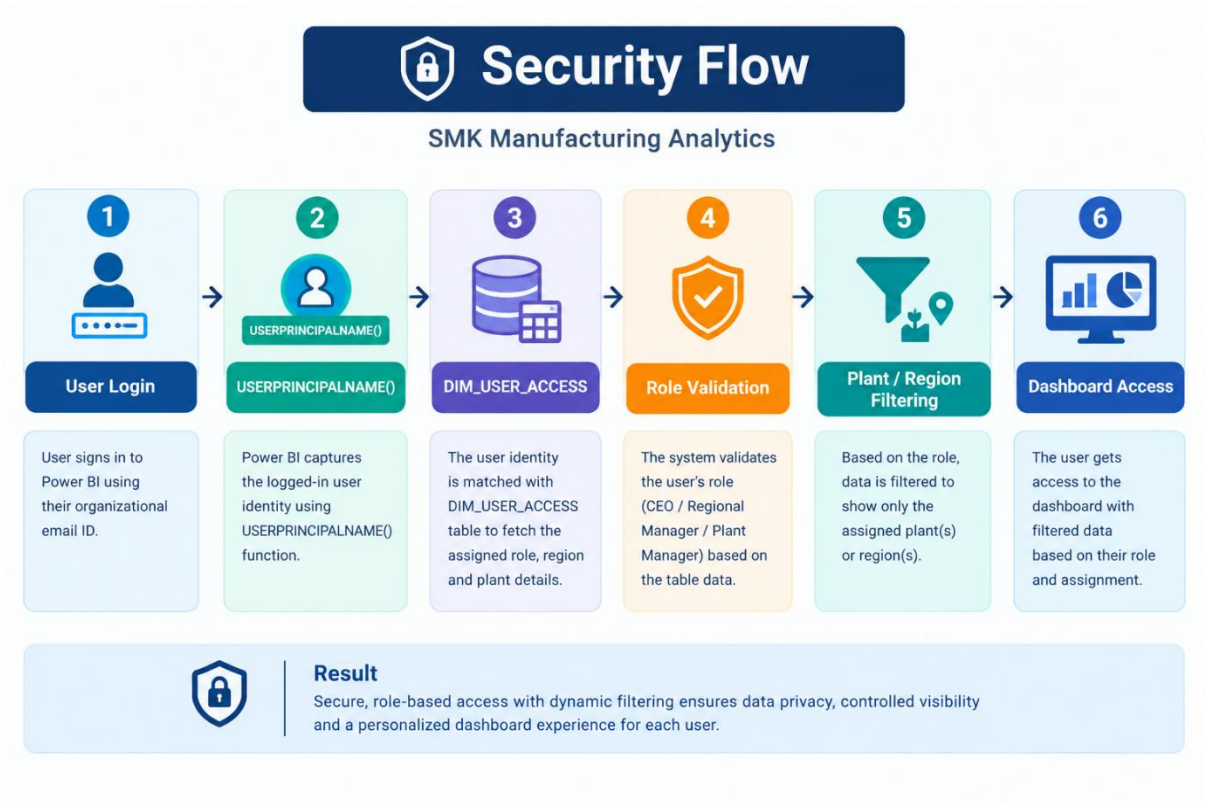
RETURN
DIM_PLANT[REGION] = UserRegion
```

User Access Table

The DIM_USER_ACCESS table was created to manage user-based security access and role assignments.

Column	Purpose
USER_EMAIL	Stores user login email ID
ROLE_NAME	Stores assigned user role
PLANT_ID	Assigns plant access
REGION	Assigns regional access
IS_ACTIVE	Indicates active/inactive users

Security Flow



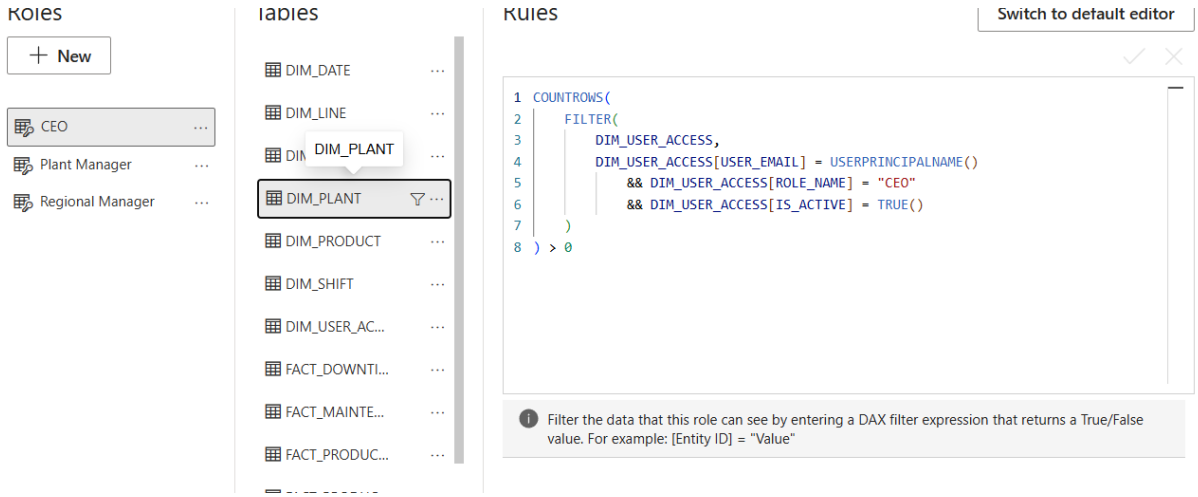
Benefits of Dynamic RLS

- Secure access control.
- Role-based data visibility.
- Scalable enterprise security model.
- Eliminates manual report maintenance.
- Supports real-world business scenarios.

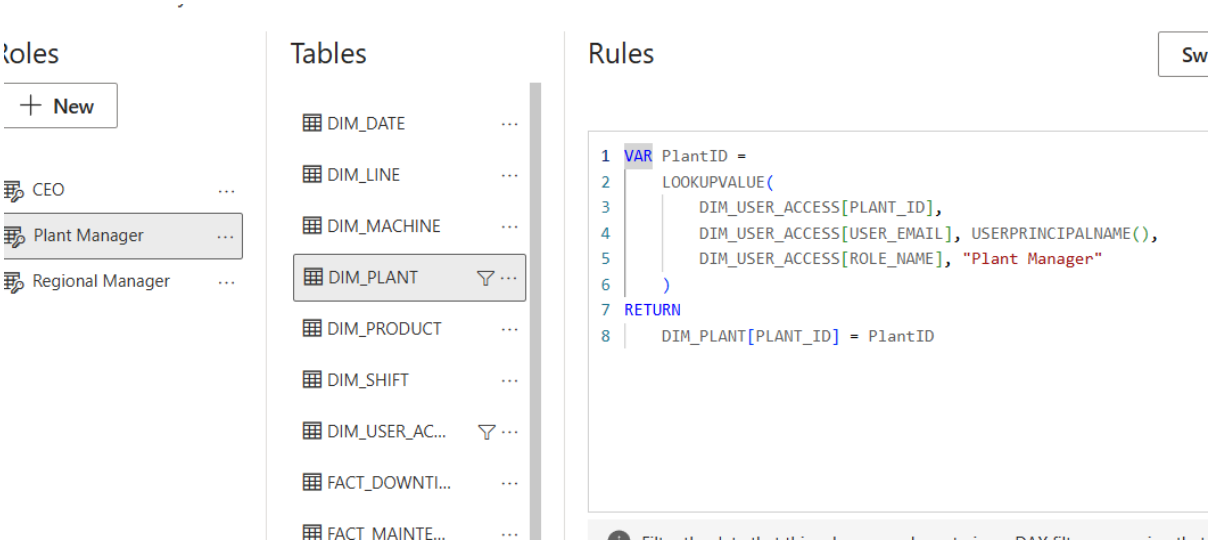
Business Value

Dynamic RLS ensures that executives, regional managers, and plant managers see only the data they are authorized to access while maintaining a single report and dataset for all users.

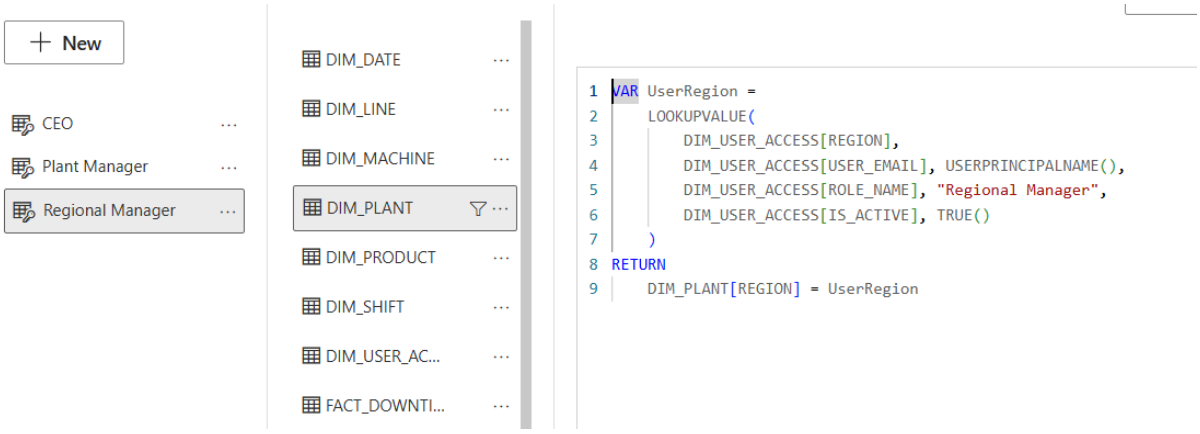
1. **CEO Role Configuration**



2. **Plant Manager Role Configuration**



3. Regional Manager Role Configuration



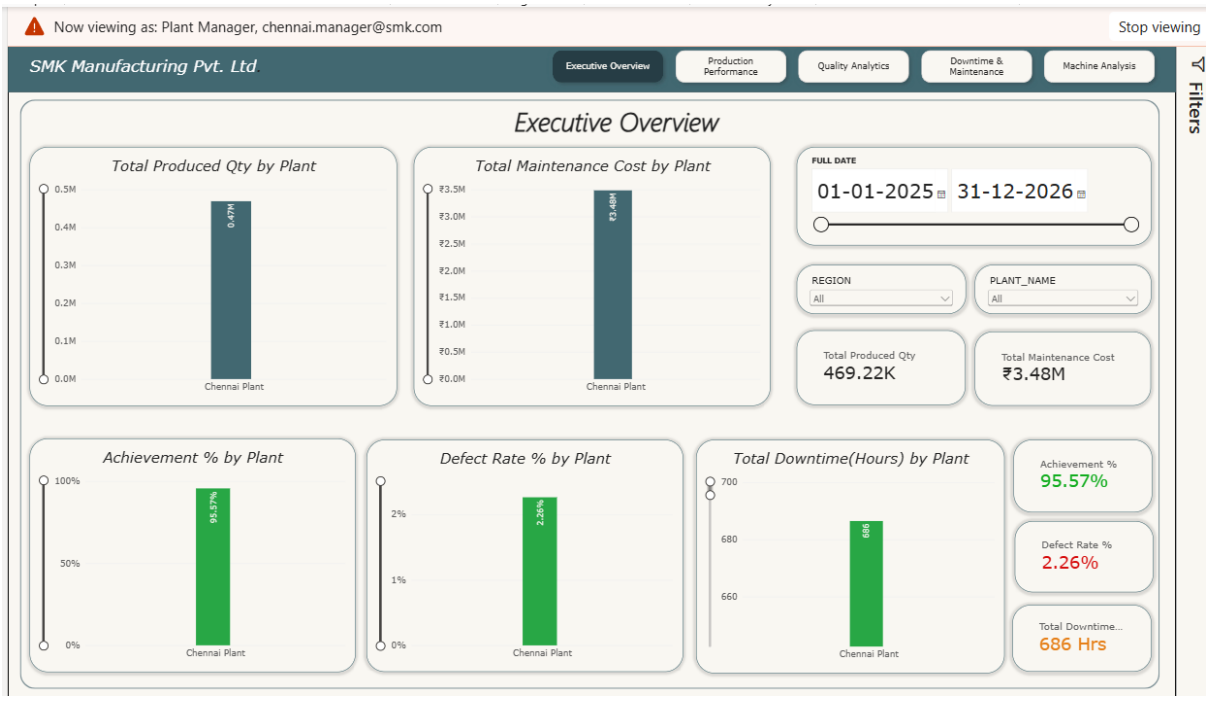
CEO Role



Regional Manager Role



Plant Manager Role



The security model uses USERPRINCIPALNAME() to dynamically filter data based on the logged-in user. Access permissions are managed through the DIM_USER_ACCESS table, enabling role-based visibility for CEO, Regional Manager, and Plant Manager users.

11. Business Insights

The SMK Manufacturing Analytics Dashboard generated several actionable insights across Production, Quality, Maintenance, Downtime, and Machine Performance. These insights help management identify operational bottlenecks and improve overall manufacturing efficiency.

Production Insights

Insight 1: Production Achievement Performance

Overall production achievement remained above target expectations, indicating stable manufacturing operations across plants.

Insight 2: Production Variance Analysis

Production variance analysis helped identify gaps between planned and actual production, enabling proactive production planning.

Insight 3: Plant Performance Comparison

Plant-level analysis highlighted the best-performing and underperforming plants based on production achievement percentage.

Quality Insights

Insight 4: Defect Rate Monitoring

Quality analysis revealed overall defect trends across plants, helping management focus on quality improvement initiatives.

Insight 5: Major Defect Categories

Defect-type analysis identified the defect categories contributing the highest number of quality issues.

Insight 6: Pass Rate Performance

Pass rate evaluation demonstrated product quality consistency across manufacturing plants.

Maintenance & Downtime Insights

Insight 7: Downtime Analysis

Plant-wise downtime analysis identified locations experiencing the highest operational interruptions.

Insight 8: Maintenance Cost Evaluation

Maintenance cost monitoring highlighted plants and machine types requiring higher maintenance expenditure.

Insight 9: Downtime Category Analysis

Category-wise downtime analysis helped identify major causes of production loss and equipment inefficiencies.

Machine Performance Insights

Insight 10: Best Performing Machine

The dashboard identifies the machine type with the highest production achievement percentage, allowing management to benchmark operational best practices.

Insight 11: Worst Performing Machine

The dashboard highlights the machine with the highest downtime, helping maintenance teams prioritize corrective actions.

Insight 12: Machine Cost Analysis

Machine-level maintenance cost analysis supports better resource allocation and maintenance planning.

Overall Business Impact

- Achieved **95.42% Production Achievement**, indicating strong production performance against planned targets.
- Maintained a low **2.35% Defect Rate**, demonstrating effective quality control processes.
- Monitored **4,118 Hours of Downtime** across manufacturing plants to improve operational efficiency.
- Identified **Grinding Machine (95.6%)** as the best-performing machine based on achievement percentage.
- Identified **CNC Machine (1,312 Hours Downtime)** as the machine requiring focused maintenance attention.
- Enabled centralized KPI monitoring and secure data access using Dynamic Row-Level Security (RLS).

12. Limitations

While the SMK Manufacturing Analytics Dashboard provides comprehensive reporting and analysis capabilities, the current implementation has the following limitations:

Data Limitations

- The project is developed using sample manufacturing data for demonstration and portfolio purposes.
- Dashboard insights are dependent on the quality and accuracy of the source data.

Technical Limitations

- No live ERP system integration is included.
- No IoT or sensor-based machine monitoring is implemented.
- Real-time streaming data is not available.
- Data refresh is dependent on scheduled refresh configurations.

Analytical Limitations

- Predictive maintenance analysis is not included.
- Forecasting and machine learning models are not implemented.

Security Limitations

- Security is implemented using Dynamic Row-Level Security (RLS) within Power BI.
- Additional enterprise security integrations such as Single Sign-On (SSO) and Azure Active Directory group-based security are not included.

Future Enhancements

- Real-time manufacturing monitoring.
- ERP system integration.
- Predictive maintenance analytics.
- Production forecasting using Machine Learning.
- Automated alert and notification system.

Summary

Despite these limitations, the solution successfully delivers centralized reporting, KPI monitoring, drill-down analysis, machine performance tracking, and secure role-based access for manufacturing operations.

13. Conclusion

The **SMK Manufacturing Analytics Dashboard** successfully delivers a centralized Business Intelligence solution for monitoring and analyzing Production, Quality, Maintenance, Downtime, and Machine Performance across multiple manufacturing plants.

By leveraging **Snowflake**, **Power BI**, **DAX**, and **Dynamic Row-Level Security (RLS)**, the solution provides a scalable and secure reporting platform that enables business users to track KPIs, monitor operational performance, and make data-driven decisions.

The dashboard offers comprehensive insights into production efficiency, quality performance, maintenance activities, downtime trends, and machine utilization through interactive visualizations and drill-down capabilities.

Key Achievements

- Developed an end-to-end Manufacturing Analytics solution.
- Implemented a Star Schema data model for efficient reporting.
- Created business-focused DAX measures and KPIs.
- Designed interactive dashboards with drill-down analysis.
- Implemented Dynamic Row-Level Security (RLS) using USERPRINCIPALNAME().
- Enabled plant-level, regional-level, and executive-level reporting.
- Delivered actionable insights to support operational decision-making.

Business Value Delivered

- Improved visibility into manufacturing operations.
- Enhanced production performance monitoring.
- Better quality and defect analysis.
- Effective maintenance and downtime tracking.
- Improved machine utilization analysis.
- Secure and role-based access to business data.

Project Outcome

The SMK Manufacturing Analytics Dashboard demonstrates practical expertise in:

- Snowflake Data Warehousing
- SQL Querying
- Data Modeling (Star Schema)
- Power BI Dashboard Development
- DAX Measure Creation
- Dynamic Row-Level Security (RLS)
- Business Performance Analytics

This project serves as a complete end-to-end Manufacturing Analytics solution and showcases the ability to transform raw data into meaningful business insights for decision-making and operational excellence.